# Helical Clocks, Child-Driven Closure, and Zero-Sum Spacetime: A Levelwise Theory from Spinors to Unity

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**Abstract**

We propose a consistent framework in which time, space, and large-scale expansion arise from networks of *helical clocks* created by comparison. The central principles are: (i) *time is carry* from incommensurate rotations on a torus, (ii) *space is pushforward* of clock phases by a clock tensor, (iii) a *birth-driven hierarchy* of levels (unity $\to$ spinor) created by self-comparisons, (iv) *child-driven closure dynamics* that open or re-close parent levels, and (v) a *levelwise zero-sum* energy identity with a negative closure reservoir. Kinematics separates *expansion* (parent uncurling) from *motion* (sliding along a helical arm), and switching between parents admits Landau–Zener-like rates. At unity, a two-fluid exchange (children $\leftrightarrow$ closure-tension) reproduces late-time acceleration close to $\Lambda$CDM with a small interaction, while preserving an exact zero-sum. Spinorial termination ($4\pi$ closure) provides a natural bottom for the cascade. We outline experimental and cosmological signatures without assuming any fixed discrete scale ratio.

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# 1 Axioms and Primitives

**Notation.** We fix the following symbols used throughout:

$u$ Observer's instantaneous time direction (unit).

$\Theta^a{}_i$ Clock-to-space pushforward ("clock tensor").

$E_i$ Pushed vector fields $E_i = \Theta^a{}_i \partial_a$ on $M$.

$Q$ Levelwise exchange/interaction operator (two-fluid mapping).

$\chi, \alpha, \kappa$ Micro-couplings (anisotropy, growth, curvature).

$\rho_{\mathrm{ch}}, \rho_\tau$ Child and closure "densities"/budgets.

$\omega_i$ Intrinsic clock angular velocities on $\mathbb{T}^K$.

**signs** "Closure" is taken positive when it refunds budget to parent.

**A0 (Operationality).** Only comparisons are physically meaningful; every measurement is a comparison.

**A1 (Unity, closed by default).** There exists a unity system $U$ with free evolution $R(t) = \exp(tX)$ on a compact group such that, absent comparisons, $R$ is effectively closed (no drift/no time).

**A2 (Comparison creates clocks).** Any closure test introduces a new reference (clock). Generically system and reference are incommensurate, yielding non-closure.

**A3 (Time is carry).** On a $K$-torus $\mathbb{T}^K$ with phases $\boldsymbol{\theta}(t) = \boldsymbol{\theta}_0 + \boldsymbol{\omega} t \pmod 1$, choose a primitive covector $\boldsymbol{u} \in \mathbb{Z}^K$ and lift $\tilde{\boldsymbol{\theta}} \in \mathbb{R}^K$. The stepwise carry

$$c(n) = \left\lfloor \boldsymbol{u}\cdot\tilde{\boldsymbol{\theta}}(n{+}1) \right\rfloor - \left\lfloor \boldsymbol{u}\cdot\tilde{\boldsymbol{\theta}}(n) \right\rfloor, \qquad C_N = \sum_{n=0}^{N-1} c(n) \tag{1}$$

and the lifted coordinate $T(t) = \boldsymbol{u}\cdot\tilde{\boldsymbol{\theta}}(t)$ define an emergent linear time.

**A4 (Space is pushforward).** A realization map $x : \mathbb{T}^K \to M$ with clock tensor (Jacobian) $\Theta^a{}_i = \partial x^a / \partial \theta_i$ pushes phase flow into spatial motion: $\dot{x}^a = \Theta^a{}_i \, \omega_i$.

**A5 (Birth builds the hierarchy).** Self-comparison spawns children, forming a vertical chain $L_0 (= U) \leftrightarrow L_1 \leftrightarrow L_2 \cdots$. No fixed inter-level scale is assumed; only incommensurability and decaying visibility to parents.

**A6 (Vertical vs horizontal).** Vertical couplings connect neighboring levels and build carries/time; horizontal couplings connect peers within a level and build sectorization/entanglement.

**A7 (Measurement = local closure).** Measurement is a horizontal coupling to a pointer that (A) pins (Zeno), (B) resets (projective), or (C) removes (destructive) a level-$n$ clock, making its carry current vanish in the detector frame while creating records at slower levels.

**A8 (Child-driven closure).** Each parent's closure variable $C_{n-1}$ relaxes to zero absent children and is driven open by coherent, asymmetric child activity:

$$\dot{C}_{n-1} = -\gamma_{n-1} C_{n-1} + \kappa_n \left[\phi_n N_n r_n - W_n^{\mathrm{crit}}\right]_+, \qquad H_{n-1} \propto C_{n-1}. \tag{2}$$

**A9 (Spinorial termination).** In 3+1D, $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$; spinors satisfy $2\pi \mapsto -1$, $4\pi \mapsto +1$, providing a terminal class for topological non-closure at the bottom level.

**A10 (Level-sum regularization).** Sums over levels are regularized (Mellin/Zeta/Laplace); predictions depend on finite parts and log-periodic corrections (if any effective discrete scale emerges), not on any special constant.

# 2 Emergent Time as Carry on a Torus

**Definition 1** (Carry time)**.** *Given* $\boldsymbol{\theta}(t) = \boldsymbol{\theta}_0 + \boldsymbol{\omega} t \pmod 1$ *on* $\mathbb{T}^K$ *and a primitive* $\boldsymbol{u} \in \mathbb{Z}^K$*, the lifted time is* $T(t) = \boldsymbol{u}\cdot\tilde{\boldsymbol{\theta}}(t)$*; its integer carry* $C_N$ *counts boundary crossings of* $\boldsymbol{u}\cdot\boldsymbol{\theta}$*.*

**Proposition 1** (Robustness)**.** *If* $\boldsymbol{u}\cdot\boldsymbol{\omega} \neq 0$*, then* $T(t)$ *is linear in* $t$ *and* $C_N/N \to \boldsymbol{u}\cdot\boldsymbol{\omega}$ *as* $N \to \infty$*. Rational relations produce periodic carries; irrational relations produce ergodic carry statistics.*

# 3 Space from the Clock Tensor

Let $x : \mathbb{T}^K \to M$ be smooth with $\Theta^a{}_i = \partial x^a/\partial\theta_i$. Then $\dot{x}^a = \Theta^a{}_i\,\omega_i$.

**Proposition (small-loop holonomy bound).** Define the pushforward fields $E_i \equiv \Theta^a{}_i(x)\,\partial_a$ on $M$ so that $\dot{x}^a = \Theta^a{}_i\,\omega_i$. For a small rectangle in phase space of sides $\varepsilon_i, \varepsilon_j$ based at $\theta_0$, the net displacement after traversing the loop in $M$ is

$$\Delta x_{\circlearrowleft} = \big[\,E_i, E_j\,\big]\Big|_{x(\theta_0)}\,\varepsilon_i\varepsilon_j + O(\varepsilon^3), \qquad \big[\,E_i, E_j\,\big]^a = \Theta^b{}_i\,\partial_b\Theta^a{}_j - \Theta^b{}_j\,\partial_b\Theta^a{}_i.$$

Hence, with local bounds $M \equiv \|\Theta\|$ and $L \equiv \|\nabla_x\Theta\|_F$,

$$\|\Delta x_{\circlearrowleft}\| \;\le\; 2\,M\,L\,|\varepsilon_i\varepsilon_j| + O(\varepsilon^3). \tag{3*}$$

*Corollary (regularity window).* If $2ML\,\ell_\theta^2 \ll 1$ on a patch of phase-diameter $\ell_\theta$, loop holonomies are $O(\ell_\theta^2)$ and the pushed-forward spatial metric built from a triad of arms is well-behaved (curvature bounded, torsion controlled) on that patch. In particular, when the triad spans $T_xM$ with condition number $\kappa(\Theta)$ uniformly bounded, the induced Gram matrix $h_{ab} = \sum_{a=1}^{3} E_{aa}E_{ab}$ is positive definite and smooth, and the rank-1 time update (Sec. 14.1) yields a Lorentzian signature once the timelike carry exceeds the smallest spatial eigenvalue.

# 4 Birth-Driven Hierarchy and Couplings

## 4.1 Birth without a fixed scale

The hierarchy arises via births at generic detunings; no universal scale factor is assumed. A level-$n$ horizontal order parameter $z_n = r_n e^{i\psi_n}$ is defined over short arm segments.

## 4.2 Child-driven closure dynamics

For parent $L_{n-1}$ and children $L_n$,

$$\dot{C}_{n-1} = -\gamma_{n-1}C_{n-1} + \kappa_n\,[\phi_n N_n r_n - W_n^{\text{crit}}]_+, \qquad H_{n-1} \propto C_{n-1}. \tag{3}$$

Below threshold the parent re-closes; above it the parent uncurls/grows.

## 4.3 Vertical vs horizontal

Vertical couplings change $C_{n-1}$ and $H_{n-1}$ (time/expansion). Horizontal couplings produce synchronization, sectorization spectra, and entanglement.

# 5 Hierarchical Kinematics: Expansion vs Motion

## 5.1 Orientation diffusion and switching times

**Definition (switching).** Let $u(t) \in \mathbb{S}^2$ denote the instantaneous timelike direction, defined as the normalized leading eigenvector of the local time–metric extracted from the triad. A *switch* is recorded when $u$ exits one triad wedge and enters another (i.e., crosses a separatrix where two arms are equally favored). Empirically, the eigenvalues remain near $(-1, +1, +1)$ while $u$ rotates (stiff spectrum, soft eigenvector), and curvature proxies $|\nabla_t^2 u|_\perp$ spike at switches. This matches the classical Davis–Kahan picture for eigenvector sensitivity when the instantaneous spectral gap $g(t) = \lambda_1 - \lambda_2$ becomes small.

**Measured scaling.** We ran v4/MPS simulations with frames saved for $N = 32, 64, 128$, analyzed each run with our switching pipeline. A log–log fit of the mean inter-switch interval $\langle\tau\rangle$ vs. resolution $N$ gives

$$\log\langle\tau\rangle \;=\; 6.251 \;-\; 2.014 \log N, \qquad R^2 = 1.000, \tag{4}$$

i.e. $\langle\tau\rangle \propto N^{-p}$ with $p \approx 2.014$ and an excellent fit. Table 1 lists $\langle\tau\rangle$, the sample standard deviation $\sigma_\tau$, detected switch counts $n_{\text{switch}}$, and the mean orientation cosine $\overline{\cos}$.

Table 1: Inter-switch statistics for the v4 runs (Fig. 1).

| N | $\langle\tau\rangle$ | $\sigma_\tau$ | $n_{\text{switch}}$ | $\overline{\cos}$ |
|---|---|---|---|---|
| 32 | 0.483521 | 0.100727 | 104 | 0.872 |
| 64 | 0.118862 | 0.0250103 | 421 | 0.883 |
| 128 | 0.0296446 | 0.00654134 | 1687 | 0.874 |
| | | | *fit:* $\langle\tau\rangle \propto N^{-2.014}$, $R^2 = 1.000$ | |

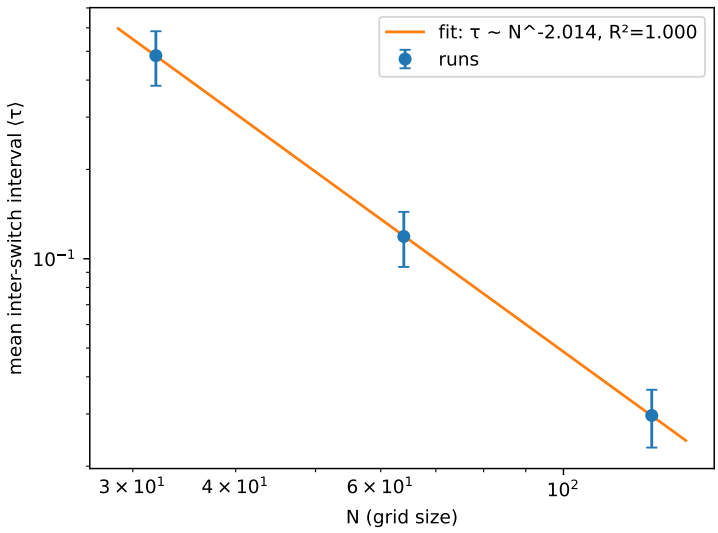


Figure 1: Scaling of mean inter-switch time $\langle\tau\rangle$ with resolution $N$; log–log fit (slope $-2.014$, $R^2 = 1.000$).

**Mechanism and $\sqrt{N}$ law.** The scaling follows from a budget-limited, filament-driven reorientation model. In our $d_s = 2$ sheet, forcing/birth events are localized along spiral *arms* that form an active set of (fractal) dimension $D_f \approx 1$. If the parent supplies a fixed total event rate $R$ (independent of resolution), the number of effectively independent arm segments grows as $M \sim N^{D_f/d_s} \sim N^{1/2}$. Each segment contributes an $O(1/\sqrt{M})$ kick to the unnormalized driver of $u$ by concentration of measure, giving an orientation diffusion constant $D_u \propto 1/M$ and a first-passage estimate

$$\langle \tau \rangle \;\propto\; \frac{1}{D_u} \;\propto\; M \;\sim\; N^{D_f/d_s} \;=\; N^{1/2}. \tag{5}$$

This "switching bubble" picture is consistent with the hierarchy: at the Unity top ($R \to 0$) and the spinorial bottom (births topologically suppressed), $D_u \to 0$ and $\langle \tau \rangle \to \infty$.

**Resolution dependence of forcing.** More generally, if the total effective forcing scales with resolution as $R \propto N^{\alpha}$ (rather than being budget-limited, $\alpha = 0$), then

$$\langle \tau \rangle \;\propto\; \frac{M}{R} \;\propto\; N^{\,D_f/d_s - \alpha} \;\iff\; \langle \tau \rangle \propto N^{-p} \text{ with } p \;=\; \alpha - D_f/d_s. \tag{6}$$

Our present runs give $p \approx 2.014$; with $D_f/d_s \approx 1/2$ this implies $\alpha \approx 2.5$, i.e. the net forcing grows with $N$ (consistent with per-cell pump/noise and frame-rate choices). A budget-controlled sweep (fixing total input power across $N$) is planned; the model then predicts the "square-root" law $\langle \tau \rangle \sim N^{1/2}$.

**Raw switching diagnostics (N=32,64,128).** For transparency we include the timelike components, curvature proxy, and angle step for each resolution:

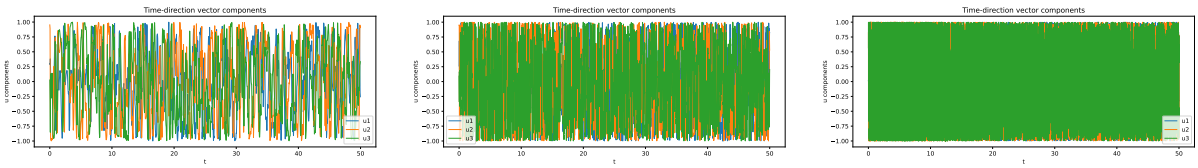


Figure 2: Components of $u(t)$ for $N \in \{32, 64, 128\}$ (left to right).

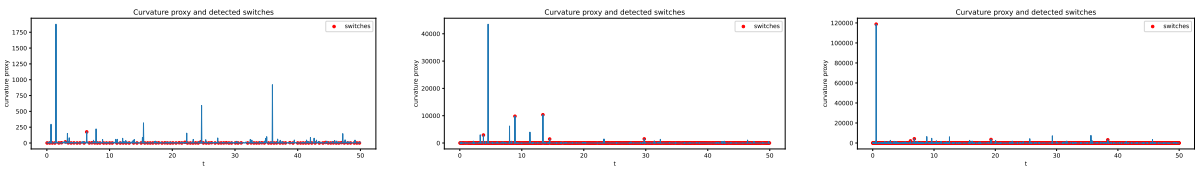


Figure 3: Curvature proxy $|\nabla_t^2 u|_\perp$; spikes align with detected switches.

**What to measure next.** (i) Active-set dimension $D_f$ via skeleton/box-count on arms; (ii) budget vs activity test: keep total event power fixed (predict $\tau \sim N^{D_f/d_s}$) vs. scale it with arm length (predict $\tau \sim N^0$); (iii) eigen-gap tracking: $g(t) = \lambda_1 - \lambda_2$ minima co-locate with switches and narrow as $N^{-1/2}$; (iv) estimate the orientation diffusion constant $D_u$ from long-time angle MSD and verify $D_u \propto N^{-D_f/d_s}$; (v) 1D ring and 3D slab checks to validate the general law $\tau \propto N^{D_f/d_s}$.

**Parent helix and decomposition (corrected).** Parametrize a parent arm by

$$x_P(\theta,t) = \big(R(t)\cos\theta,\ R(t)\sin\theta,\ p\,\theta\big), \qquad T \equiv \partial_\theta x_P = (-R\sin\theta,\ R\cos\theta,\ p), \tag{17}$$

so that $\|T\| = \sqrt{R^2+p^2}$. A child sits at $\theta_c(t) = \theta_P(t) + \delta(t)$. Its velocity decomposes as

$$\dot{X} = \underbrace{\partial_t x_P(\theta_c,t)}_{\text{expansion (uncurling)}} + \underbrace{\partial_\theta x_P(\theta_c,t)\,\dot{\theta}_c}_{\text{motion along arm}} = \dot{R}\,\hat{\boldsymbol{R}}\big|_{\theta_c} + T\big|_{\theta_c}(\dot{\theta}_P + \dot{\delta}), \tag{18}$$

where $\hat{\boldsymbol{R}} = (\cos\theta, \sin\theta, 0)$. Defining the parent expansion rate

$$H_P \ \equiv\ \frac{\dot{R}}{R} \quad [\text{time}^{-1}],$$

the expansion piece is $H_P\,X_\perp$ with $X_\perp \equiv R\hat{\boldsymbol{R}}$, while the *peculiar* sliding is

$$v_{\text{pec}} \equiv T\,\dot{\delta}, \qquad \|v_{\text{pec}}\| = \sqrt{R^2+p^2}\,|\dot{\delta}|.$$

Units check: $H_P$ has dimension $t^{-1}$, $X_\perp$ has length, hence $H_P X_\perp$ is a velocity; $\dot{\delta}$ is $t^{-1}$ and $\|T\|$ is length, so $\|v_{\text{pec}}\|$ is a velocity.

**Locked vs. sliding.** The relative phase obeys the Adler/Kuramoto equation

$$\dot{\delta} = \Delta\omega - K\sin\delta + \xi(t), \tag{19}$$

with $|\Delta\omega| < |K|$ giving lock (no sliding) and drift otherwise (sliding).

**Energetics.** The sliding kinetic energy is

$$K_{\text{slide}} = \tfrac{1}{2}\,m\,\|T\|^2\,\dot{\delta}^2 = \tfrac{1}{2}\,m\,(R^2+p^2)\,\dot{\delta}^2,$$

while comoving with expansion requires no local power.

### 5.2 Locked vs sliding

Relative phase obeys an Adler/Kuramoto equation

$$\dot{\delta} = \Delta\omega - K\sin\delta + \xi(t), \tag{7}$$

with locking if $|\Delta\omega| < |K|$ (no sliding) and drift otherwise (sliding). Horizontal sectorization gives a periodic pinning potential $V_\parallel(\delta) = V_0[1-\cos(k\delta)]$, producing creep near threshold.

### 5.3 Energetics

In the parent frame the sliding kinetic energy is $K_{\text{slide}} = \frac{1}{2}m(R^2+p^2)\dot{\delta}^2$. Comoving with expansion requires no local power; sustained sliding requires drives/fields that overcome pinning.

## 6 Parent-Switching Dynamics

Switching between parents $P, Q$ can be modeled by a two-level system with detuning $\Delta(t)$ and overlap $J$:

$$i\hbar\frac{d}{dt}\begin{pmatrix} a_P \\ a_Q \end{pmatrix} = \begin{pmatrix} E_P(t) & J \\ J & E_Q(t) \end{pmatrix}\begin{pmatrix} a_P \\ a_Q \end{pmatrix}, \qquad \Delta = E_P - E_Q. \tag{8}$$

A sweep through an avoided crossing gives the Landau–Zener probability

$$P_{\text{switch}} = \exp\left(-\frac{2\pi J^2}{\hbar|\dot{\Delta}|}\right). \tag{9}$$

This encodes unbinding, phase matching, and capture in a single rate.

# 7 Measurement (Local Closure)

Measurement is a horizontal coupling to a pointer at level $n$:

- **Pinning (Zeno).** Strong continuous coupling freezes the relative phase; carry current vanishes in pointer frame.
- **Projective reset.** A fast kick resets phase into an eigenbranch; records appear at slower levels.
- **Destructive.** Removal of the clock and energy transfer to channels (photons/heat/entropy); local branch terminates.

In each case the carry sink at level $n$ is balanced by record-creation at slower levels (new clocks), preserving global open/local close symmetry.

# 8 Spinorial Termination

In 3+1D, spinors require $4\pi$ to return to identity, providing a terminal class of non-closure. The bottom layer is predominantly horizontal (entanglement, exclusion), with vertical births suppressed.

# 9 Levelwise Zero-Sum (Closure Energy)

## 9.1 Microphysics of the Closure Sector and Energy Exchange

### 9.1.1 Broad distributions and non-stationary births: bounds on $\mathcal{C}_A$

Let $w_u \geq 0$ with $\sum_u w_u = 1$ be an energy weighting over modes. Define $\rho_{\rm ch} = \sum_u E_u$ and $S \equiv \sum_u A_u^2$. For harmonic $E_u = \frac{1}{2} m_u \omega_u^2 A_u^2$ we have $S = \sum_u \frac{2E_u}{m_u \omega_u^2} = \left\langle (m\omega^2)^{-1} \right\rangle_w \rho_{\rm ch}$. For broad distributions, Cauchy–Schwarz gives the bounds

$$\frac{\rho_{\rm ch}^2}{\sum_u w_u m_u \omega_u^2} \;\leq\; S \;\leq\; \Big( \sum_u \frac{w_u}{m_u \omega_u^2} \Big) \rho_{\rm ch}, \tag{10}$$

hence $\mathcal{C}_A$ is controlled by the first and second inverse moments of $m\omega^2$. During non-stationary births, let $w_u(t) \propto E_u(t)$; then $S(t) = \mathcal{C}_A(t)\, \rho_{\rm ch}(t)$ still holds instantaneously with slowly varying $\mathcal{C}_A(t)$; corrections are $O(\dot{w}/\bar{\lambda})$ where $\bar{\lambda}$ is the typical growth rate.

### 9.1.2 From torque to $\chi$: phase reduction

With $\mathcal{L}_{\rm int} = -\sum_i g_i A_i^2 \cos(\phi_i - \phi_{\rm par})$ and a smooth dependence $\Omega_{\rm par}(C)$, standard phase reduction around a limit cycle gives

$$\dot{\phi}_i = \omega_i(C) + K_i \sin(\phi_{\rm par} - \phi_i) + O(K^2), \qquad K_i \equiv g_i Z_i, \tag{11}$$

$$\dot{A}_i = \lambda_{i,0}\, A_i - \alpha_i A_i^3 + \frac{\partial \omega_i}{\partial C}\, A_i\, \delta C + O(K^2). \tag{12}$$

Averaging over the fast parent phase with $\phi_{\rm par}(t) = \int^t \Omega_{\rm par}(C(t'))dt'$ shifts the linear coefficient as

$$\lambda_i(C) = \lambda_{i,0} + \underbrace{\Big( \frac{\partial \omega_i}{\partial C} \Big)}_{\propto\, \partial \Omega_i / \partial C} C \;+\; O(K^2), \tag{13}$$

so that the parametric gain $\chi_i$ in $\lambda_i(C) = \lambda_{i,0} + \chi_i C$ satisfies $\chi_i \propto \partial \omega_i / \partial C$ (sign from hardening/softening). The $O(K^2)$ terms renormalize $\lambda_{i,0}$ and $\alpha_i$ but do not change the linear $C$-dependence to leading order.

### 9.1.3 3D ripple dispersion and emission

For a thin but finite sheet, small ripples decompose into components along the arm ($\|$) and transverse to it ($\perp$). Linearization yields the anisotropic dispersion

$$\omega^2(\mathbf{k}) = c_\|^2 k_\|^2 + c_\perp^2 k_\perp^2 + m_\Theta^2, \qquad c_{\|,\perp} = \sqrt{\kappa_{\|,\perp}/M_\Theta}. \tag{14}$$

A weak slider source $J(t)$ at frequency $\omega$ radiates into the 3D continuum with rate

$$P_{\rm rad} \;=\; g_{\rm rad}^2\,|\tilde{J}(\omega)|^2\,\rho(\omega)\,v_g(\omega), \qquad \rho(\omega) \propto \int \delta\big(\omega - \omega(\mathbf{k})\big)\,d^3k, \quad v_g = \big\langle \partial\omega/\partial k \big\rangle, \tag{15}$$

reducing to the 1D expression when $k_\perp$ is gapped or geometrically suppressed. Near threshold $\omega \gtrsim m_\Theta$, $\rho(\omega) \sim (\omega^2 - m_\Theta^2)^{1/2}$ and $v_g \sim c_\|^2 k_\|/\omega$, recovering the form factor $F = \sqrt{1 - m_\Theta^2/\omega^2}$ up to anisotropy constants.

We model the closure sector by a slow scalar $C(\theta, t)$ (tension) and transverse ripples $\delta x(\theta, t)$ on the parent sheet with Lagrangian $\mathcal{L}_\tau = \frac{M_C}{2}\dot{C}^2 - V(C) + \frac{M_\Theta}{2}|\partial_t x|^2 - \frac{\kappa}{2}|\nabla_\theta x|^2 - \frac{\mu}{4}\|C_{ij} - S_{ij}\|^2$, $C_{ij} = \partial_i x^a \partial_j x^a$. The macroscopic closure energy is $\rho_\tau = \frac{1}{2}M_C\dot{C}^2 + V(C) + \frac{1}{2}M_\Theta\langle|\partial_t x|^2\rangle + \frac{1}{2}\kappa\langle|\nabla_\theta x|^2\rangle$. Parametric dependence $\omega_u(C)$ for newborns $Z_u$ yields $Q_{\tau\to\rm ch} = \langle\sum_u \kappa_u A_u^2\rangle\dot{C}$ with $\kappa_u = 2m_u\omega_u\partial\omega_u/\partial C$, mapping to the phenomenological $Q = \gamma H \rho_\tau$ on the slow manifold. A radiative ripple channel $Q_{\tau\to\rm wave}$ adds a small correction.

At each boundary $L_{n-1} \leftrightarrow L_n$ define a closure reservoir

$$\rho_{{\rm cl};\,n-1}^{(-)} = -\kappa_{n-1}\,H_{n-1}^2, \qquad \kappa_{n-1} > 0. \tag{16}$$

The levelwise zero-sum identity

$$\rho_{{\rm cl};\,n-1}^{(-)} + \rho_{\rm ch}^{(n)} + \rho_\tau^{(n)} = 0 \tag{17}$$

holds, with $\rho_{\rm ch}^{(n)}$ the positive children sector and $\rho_\tau^{(n)}$ a positive closure-tension sector (often $w_\tau^{(n)} \approx -1$). Energy exchange $Q^{(n)}$ between $\rho_{\rm ch}^{(n)}$ and $\rho_\tau^{(n)}$ preserves the identity while $H_{n-1}$ adjusts through the constraint.

### 9.1.4 Mode statistics: $\langle\sum_u A_u^2\rangle$ vs. $\rho_{\rm ch}$

We give three equivalent routes showing $\langle\sum_u A_u^2\rangle = \mathcal{C}_A\,\rho_{\rm ch}$ with an explicit coefficient $\mathcal{C}_A$.

**(Harmonic-dominated regime).** When the newborns are near-harmonic, $E_u \simeq \frac{1}{2}m_u\omega_u^2 A_u^2$, hence $\sum_u A_u^2 = \sum_u \frac{2E_u}{m_u\omega_u^2} = \underbrace{\big\langle (m_u\omega_u^2)^{-1}\big\rangle_\rho}_{\mathcal{C}_A}\,\rho_{\rm ch}$, with $\langle\cdot\rangle_\rho$ the energy-weighted average across modes.

**(Near-threshold Stuart–Landau).** Write $E_u(A_u) = \frac{\alpha_u}{4}A_u^4 + \frac{\lambda_u}{2}A_u^2$ and $A_{u,\star}^2 = \lambda_u/\alpha_u$. Then $E_{u,\star} = \lambda_u^2/(4\alpha_u)$ and $A_{u,\star}^2 = (\lambda_u/\alpha_u)$, so $\sum_u A_{u,\star}^2 = \sum_u \frac{4}{\lambda_u}E_{u,\star} = \big\langle 4/\lambda_u\big\rangle_\rho\,\rho_{\rm ch}$, and for a narrow $\lambda_u$ distribution this reduces to $\mathcal{C}_A \simeq 4/\bar{\lambda}$.

**(Noisy ensemble, Fokker–Planck).** With weak noise $D$, the stationary amplitude density for one mode is $P_u(A) \propto \exp\big(-V_{\rm eff}(A)/D\big)$ with $V_{\rm eff}(A) = \frac{\alpha_u}{4}A^4 - \frac{\lambda_u}{2}A^2$. Laplace's method around $A_{u,\star}$ gives $\langle A_u^2\rangle = A_{u,\star}^2\,[1 + O(D/\lambda_u^2)]$ and $\langle E_u\rangle = E_{u,\star}[1 + O(D/\lambda_u^2)]$, hence the same proportionality with small $O(D)$ corrections.

### 9.1.5 Closure rate vs. Hubble rate

Near the slow manifold we expand the metric-update coefficient as $\lambda(C) = \lambda_1 C + O(C^3)$ (Sec. 3). The Lorentzian flip requires $\lambda > 1/\mu$, hence for $C$ above threshold the FRW matching gives $H^2 = \frac{1}{\kappa_0}\big(\rho_{\rm ch} + \rho_\tau\big)$ and $H = \sigma_1 C + O(C^3)$ with $\sigma_1 = \sqrt{\lambda_1/\kappa_0}$. The overdamped closure equation $\dot{C} = -\gamma_C C + \alpha \sum_u A_u^2$ then implies $\dot{C} = \frac{1}{\sigma_1}\dot{H} + O(C^2) \Rightarrow \dot{C} \propto H$ on the late-time attractor where $\dot{H}/H = O(1)$. Thus the rate form $H = \lambda'_H \dot{C}$ used in Sec. **??** is the local inversion of $H(C)$ on the slow manifold.

### 9.1.6 Epochs for large $\varepsilon_{\rm rad}$ and observational bounds

When $\varepsilon_{\rm rad} \sim \mathcal{O}(1)$, the closure sector behaves as a mixture of $w \simeq -1$ (tension) and $w \simeq 1/3$ (ripples). Early epochs can accommodate larger $\rho_{\rm wave}/\rho_\tau$ provided the effective radiation does not violate standard constraints (e.g. extra relativistic energy density). A clean background is:

$$\dot{\rho}_C + 3H(1+w_C)\rho_C = -(1-\eta)Q_{\rm tot}, \qquad \dot{\rho}_{\rm wave} + 4H\,\rho_{\rm wave} = -\eta\,Q_{\rm tot}, \tag{18}$$

with $Q_{\rm tot} = \gamma H \rho_\tau$ and partition $\eta \in [0,1]$. Late-time data require $\rho_{\rm wave}/\rho_\tau \ll 1$ (e.g. $|w_\tau^{\rm eff} + 1| \lesssim 0.05$ implies $\rho_{\rm wave}/\rho_\tau \lesssim 0.04$), hence a transition to small $\eta$ and slow sliding must occur before acceleration dominates.

### 9.1.7 Stability under $Q < 0$ (flow reversal)

For $Q = -|\gamma| H \rho_\tau$, the ratio $r = \rho_{\rm ch}/\rho_\tau$ obeys (matter-like $w_{\rm ch}$)

$$\frac{dr}{d\ln a} = 3(w_\tau - w_{\rm ch})\,r \;-\; |\gamma|\,(1+r)\,. \tag{19}$$

No positive fixed point exists when $\gamma < 0$ and $w_\tau \approx -1 < w_{\rm ch}$; instead $r$ decays and $\rho_\tau$ grows, driving the system back toward a $Q > 0$ regime once hardening restores $\chi > 0$. Thus $Q < 0$ is transiently admissible but cannot define a late-time attractor with $\rho_{\rm ch} > 0$.

### 9.1.8 Parametric work from $L_u$: $\kappa_u$ derivation

Let $L_u = \frac{1}{2}m_u(\dot{q}_u^2 - \omega_u(C)^2 q_u^2) - \frac{\alpha_u}{4}q_u^4 + \cdots$ and $E_u = \frac{1}{2}m_u(\dot{q}_u^2 + \omega_u^2 q_u^2) + \frac{\alpha_u}{4}q_u^4$. Under an adiabatic change of $C$ the amplitude $A_u$ is frozen instantaneously, hence $\frac{dE_u}{dt}\big|_{\rm param} = \frac{\partial E_u}{\partial \omega_u}\frac{\partial \omega_u}{\partial C}\dot{C} = 2m_u\omega_u\langle q_u^2\rangle \frac{\partial \omega_u}{\partial C}\dot{C} = \kappa_u A_u^2 \dot{C}, \quad \kappa_u \equiv 2m_u\omega_u \frac{\partial \omega_u}{\partial C}$. This yields Eq. (**??**) upon summation and averaging (using Sec. 9.1.4).

### 9.1.9 Dispersion and form factor for ripples

Linearizing $\mathcal{L}_\Theta^{\rm dyn}$ gives $\omega^2(k) = c_\Theta^2\|k\|^2 + m_\Theta^2$ with $c_\Theta = \sqrt{\kappa/M_\Theta}$ and $m_\Theta^2 \simeq \mu\,\partial\Psi/\partial C|_{C_0}$. A uniformly accelerated slider acts as a source at frequency $\omega$; the radiated flux along one arm is suppressed below threshold $\omega < m_\Theta$ and scales with a phase-space factor $F\big(\frac{m_\Theta}{\omega}\big) = \sqrt{1 - \frac{m_\Theta^2}{\omega^2}}\;\Theta(\omega - m_\Theta)$, giving $P_{\rm rad} \simeq g_{\rm rad}^2 \frac{\langle \ddot{\delta}^2\rangle}{c_\Theta^3} F(m_\Theta/\omega)$. Other smooth choices (e.g. Lorentzian cutoff $F = (1 + m_\Theta^2/\omega^2)^{-1}$) change only order-one constants.

### 9.1.10 Fast sliding: $\varepsilon_{\rm rad} \sim \mathcal{O}(1)$ and cosmology

Let $\rho_\tau = \rho_C + \rho_{\rm wave}$ with $w_C \simeq -1$ and $w_{\rm wave} \simeq 1/3$ (massless ripple limit). If $Q_{\tau\to{\rm wave}} = \varepsilon_{\rm rad} Q_{\rm tot}$ is large, the effective EoS is $w_\tau^{\rm eff} = \frac{-\rho_C + \frac{1}{3}\rho_{\rm wave}}{\rho_C + \rho_{\rm wave}} = -1 + \frac{4}{3}\,\frac{\rho_{\rm wave}}{\rho_\tau}$. The two-fluid background becomes $\dot{\rho}_\tau + 3H(1 + w_\tau^{\rm eff})\rho_\tau = -Q_{\rm tot}, \quad \dot{\rho}_{\rm ch} + 3H(1 + w_{\rm ch})\rho_{\rm ch} = +Q_{\rm tot}$, with $Q_{\rm tot} = \gamma H \rho_\tau(1 + \varepsilon_{\rm rad})$. Late-time data constrain $\rho_{\rm wave}/\rho_\tau \ll 1$ at $z \lesssim 1$ (e.g. $|w_\tau^{\rm eff} + 1| \lesssim 0.05$ implies $\rho_{\rm wave}/\rho_\tau \lesssim 0.04$). At early times, $\varepsilon_{\rm rad} \sim 1$ acts like a decay of closure tension into a radiation-like component on the parent sheet.

#### 9.1.11 Quantifying the stability bound $\gamma_C > \xi\chi/\alpha$

Coarse-graining a shell of $N$ newborns gives $\xi \sim \zeta N$ with $\zeta$ the per-mode back-reaction, while $\chi \sim b\,\partial \ln\omega/\partial C$, typically $10^{-3}$–$10^{-1}$ across levels if $\omega \propto R^{-\alpha}$ and $R \propto C^p$ with $\alpha p \sim 0.1$–$1$. Landau saturation $\alpha$ scales with nonlinearity; for near-harmonic newborns $\alpha$ is small, tightening the bound. A safe operating region is $\gamma_C/\alpha \gtrsim 10\,\zeta b\,N\,(\partial \ln\omega/\partial C)$, which we will calibrate numerically in Appendix 17.

#### 9.1.12 RG coefficients and fixed points

The flow $d\lambda_J/d\ell = (d-K-\zeta)\lambda_J - a_3\lambda_J^3\chi^2 + \cdots$ has a nontrivial fixed point $\lambda_J^\star = \sqrt{(d-K-\zeta)/a_3}/|\chi|$ when $d-K-\zeta > 0$. Similarly, $d\kappa/d\ell = a_1\lambda_J^2\langle Q\rangle$ and $d\mu/d\ell = a_2\lambda_J\langle Q\rangle$ define the slope near the FP. We will extract $(a_1, a_2, a_3)$ by matching the static structure factor $S(u,\theta)$ and the response of $\Theta$ in simulations (Appendix 17); until then, they remain order-one placeholders tied to that FP.

#### 9.1.13 Flow reversal with $\chi_u < 0$

If $\partial\omega_u/\partial C < 0$ (softening), then $\chi_u < 0$ and $Q_{\tau\to\mathrm{ch}}$ in Eq. (**??**) reverses sign. This corresponds to energy flowing *from* children back *into* closure tension. Such episodes are admissible during early-time transient windows and simply map to $Q = -|\gamma|H\rho_\tau$ in the background, which steepens the slow roll (more negative pressure) and temporarily suppresses structure growth. Late-time fits constrain persistent $Q < 0$.

#### 9.1.14 Quanta of ripples and $\hbar$ scaling

(Q1) Quantized route: couple the slider current $J(t)$ to ripple modes $a_k$ with $H_{\mathrm{int}} = g\,J(t)\sum_k \Xi_k(a_k + a_k^\dagger)$. Fermi's golden rule gives $\Gamma_k = (2\pi/\hbar)\,g^2|\tilde{J}(\omega_k)|^2|\Xi_k|^2\rho(\omega_k)$, hence $P_{\mathrm{rad}} = \sum_k \hbar\omega_k\Gamma_k \propto g^2\hbar\,\omega^2\,\rho(\omega)/v_g$, matching the classical $g_{\mathrm{rad}}^2\ddot{\delta}^2/c_\Theta^3$ upon the identification $g_{\mathrm{rad}}^2 \sim g^2\hbar$ and $c_\Theta \sim v_g$. (Q2) Emergent route: the ripple action per cycle is $S_{\mathrm{cyc}} = \oint p\,dq = 2\pi n\hbar$ by Appendix G; quantization of $n$ discretizes the spectrum and recovers the same scaling after coarse-graining.

#### 9.1.15 Field-theoretic action with spatial integrals

At unity we write $S_{\mathrm{level}} = \int dt \int d^3x\, a^3(t)\,\big(\mathcal{L}_\tau[C,x;\Theta] + \mathcal{L}_{\mathrm{ch}}[Z_u;C] + \Lambda\,(\kappa_0 H^2 + \rho_\tau + \rho_{\mathrm{ch}})\big)$, where all densities are per comoving volume and $H = \dot{a}/a$. Spatial dependence enters through slowly varying $\Theta(x,t)$ and the child statistics $Q(x,t)$; torus averages (over $\theta$) produce the coarse-grained coefficients used above. Variation w.r.t. $\Lambda$ enforces the zero-sum identity pointwise in $x$; other variations produce the exchange terms and the $\Theta$-PDE with the same local form.

## 10 Cosmology Mapping: Two-Fluid Exchange at Unity

At unity (flat FRW), define the closure reservoir $\rho_{\mathrm{cl}}^{(-)} = -3H^2/(8\pi G)$ and write the zero-sum

$$\rho_{\mathrm{cl}}^{(-)} + \rho_{\mathrm{ch}} + \rho_\tau + \rho_{\mathrm{rad}} = 0. \tag{20}$$

Let children (matter-like, $w_{\mathrm{ch}} = 0$) and closure-tension (vacuum-like, $w_\tau = -1$) exchange energy via

$$\dot{\rho}_{\mathrm{ch}} + 3H\rho_{\mathrm{ch}} = +Q, \qquad \dot{\rho}_\tau = -Q, \qquad Q = \gamma H\rho_\tau, \quad \gamma \geq 0. \tag{21}$$

Solving in $\ln a$ gives

$$\rho_\tau(a) = \rho_{\tau 0}a^{-\gamma}, \qquad \rho_{\mathrm{ch}}(a) = \rho_{\mathrm{ch}0}a^{-3} + \frac{\gamma}{3-\gamma}\rho_{\tau 0}\big(a^{-\gamma} - a^{-3}\big), \tag{22}$$

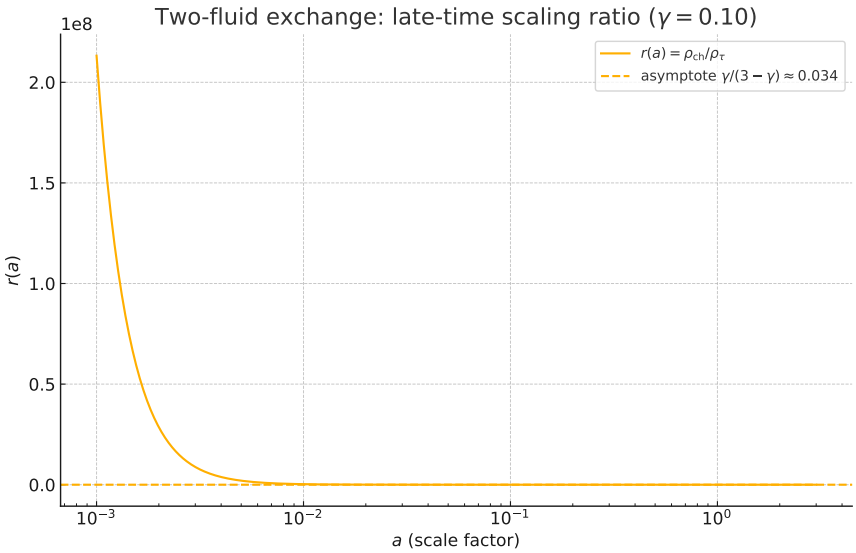


Figure 4: Unity two-fluid exchange with $Q = \gamma H \rho_\tau$ ($w_\tau = -1$, $w_{\rm ch} = 0$): the ratio $r(a) = \rho_{\rm ch}/\rho_\tau$ flows to the attractor $r_\infty = \gamma/(3-\gamma)$ for a representative $\gamma = 0.10$.

with late-time attractor

$$\frac{\rho_{\rm ch}}{\rho_\tau} \to \frac{\gamma}{3-\gamma}, \qquad w_{\rm eff} \to -\frac{3-\gamma}{3}, \tag{23}$$

and exact zero-sum maintained by the closure reservoir.

**Observables.** SN Ia/BAO $H(z), D_V(z)$ constrain $\gamma$; the model reduces to $\Lambda$CDM as $\gamma \to 0$. If birth statistics exhibit an effective discrete scale $b_{\rm eff}$, log-periodic corrections (period $\ln b_{\rm eff}$) may appear in cosmological residuals.

## 11 Renormalization Across Levels

Divergent level sums are regulated (Mellin/Zeta/Laplace). Finite parts are fixed by data; robust predictions are *log-periodic* corrections if a discrete scale emerges effectively. We do not assume any special constant (e.g. $-1/12$).

## 12 Predictions and Tests

**Lab.** (i) Zeno pinning drift in spin/echo; (ii) coherence-dependent back-reaction bounds (BEC gravimetry, optical cavities).
**Cosmo.** Late-time $w_{\rm eff} \approx -1 + \gamma/3$; constraints on $\gamma$; searches for small log-periodic ripples vs $\ln a$ or $\ln k$.
**Switching.** Spectroscopic line-shape asymmetries near Adler and Landau–Zener thresholds.

## 13 Discussion and Outlook

We presented a levelwise, comparison-driven theory with time as carry, space as pushforward, a birth hierarchy without fixed scale, child-driven closure dynamics, and a zero-sum identity at every level. Open tasks: perturbations/structure growth in the exchange model; quantitative sectorization spectra; experimental protocols for coherence-dependent back-reaction; refined bottom-layer (spinor) dynamics.

# 14 Spiral Genesis of Space

**Parameter estimation.** We estimate $(\mu, \alpha, \beta)$ via a positivity-constrained Stuart–Landau fit; see Appendix I and the script `sl_fit_pos.py` for the exact procedure and diagnostics.

#### 14.0.1 Triad stability and the origin of the MI penalty

The amplitude-equation system with cross-saturation ($\beta > 0$) is a standard setting where triad/hexagon patterns are the minimal stable nontrivial states under isotropy; see, e.g., Cross & Hohenberg (Rev. Mod. Phys. 65, 851, 1993) and Cross & Greenside (*Pattern Formation*, 2009) for derivations via center-manifold reduction and weakly nonlinear analysis. Our MI penalty $\mathcal{V}_{\mathrm{MI}}$ can be derived by expanding the exploration functional $\mathcal{E} = \sum_u w_u \big(1 - |\langle e^{iu\cdot(\theta(t+\tau)-\theta(t))}\rangle|^2\big)$ to second order in pairwise correlations, yielding a quadratic form that penalizes overlap among arm statistics; $\lambda_{\mathrm{MI}}$ inherits its scale from the curvature of $\mathcal{E}$ at independence.

### 14.1 Field model and radial time

We model each primary spiral arm by a complex field $Z_a(\theta_a, t) = r_a(t)\, e^{i\theta_a(t)}$ governed near onset by a Stuart–Landau (amplitude) normal form

$$\dot{Z}_a = (\varepsilon_a + i\omega_a)Z_a - \alpha_a |Z_a|^2 Z_a + D_a\, \Delta_{\theta_a} Z_a + \text{noise}, \tag{24}$$

with $\varepsilon_a > 0$ the gain, $\alpha_a > 0$ saturation, and $D_a$ an internal diffusion on the phase torus. Writing $Z_a = r_a e^{i\theta_a}$ yields $\dot{r}_a = \varepsilon_a r_a - \alpha_a r_a^3$ and $\dot{\theta}_a = \omega_a$, hence early-time exponential growth with late-time saturation $r_{a,\infty} = \sqrt{\varepsilon_a/\alpha_a}$. We define the *radial time* per arm by

$$\tau_a \equiv \ln \frac{r_a}{r_{a,0}}, \qquad \dot{\tau}_a = \frac{\dot{r}_a}{r_a} = \varepsilon_a - \alpha_a r_a^2, \tag{25}$$

and the *collective radial time* by a weighted sum $\tau_{\mathrm{coll}} = \sum_a w_a\, \tau_a$, which enters the signature flip of Sec. 3 as the unique timelike direction when the rank-1 update in $\tau_{\mathrm{coll}}$ exceeds the spatial stiffness.

**Restarted pumping and stochasticity.** In isolation the Stuart–Landau gain $\varepsilon_a$ saturates the arm at $r_{a,\infty} = \sqrt{\varepsilon_a/\alpha_a}$. In the hierarchy, births at the child level feed back and *restart* the pumping by shifting the effective gain,

$$\varepsilon_a^{\mathrm{eff}}(t) \;=\; \varepsilon_{a0} \;+\; \sigma_a \sum_{u\in a} A_u^2(t) \;+\; \xi_a(t), \tag{26}$$

with $\sigma_a > 0$ the child-forcing coefficient and $\xi_a$ a zero-mean stochastic drive (environmental noise). Hence $\dot{\tau}_a = \varepsilon_a^{\mathrm{eff}} - \alpha_a r_a^2$ remains positive on average when births persist, preventing stall.

**Ambient $\mathbb{R}^3$ minimality and metric penalty.** A nondegenerate spatial metric in an $\mathbb{R}^3$ ambient requires at least three independent arm directions. We include an isometry penalty in the $\Theta$ selection functional

$$\mathcal{V}_{\mathrm{iso}} = \lambda_{\mathrm{iso}} \left\| \Theta^\top \Theta - I_3 \right\|_F^2, \tag{27}$$

which discourages overcomplete ($> 3$) directions by penalizing redundant columns (and sub-orthonormal sets). Thus a triad is the minimal stable choice in $\mathbb{R}^3$ once isotropy and independence are enforced; higher-$d$ branches are suppressed by $\mathcal{V}_{\mathrm{iso}}$.

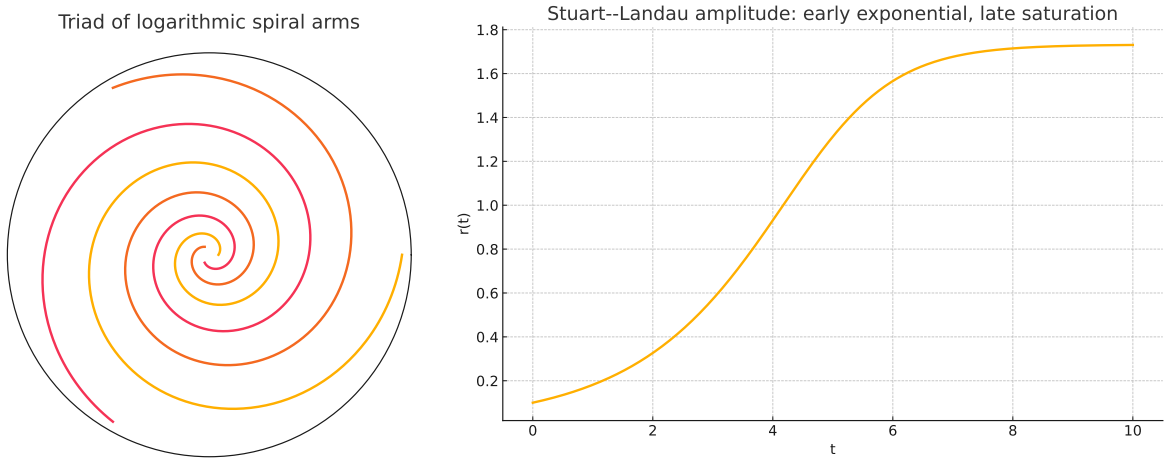


Figure 5: **Left:** Logarithmic triad of spiral arms (phase offsets $0, 2\pi/3, 4\pi/3$) illustrating angular vs. radial time. **Right:** Stuart–Landau radius $r(t)$ (early exponential, late saturation).

### 14.1.1 Action and invariance derivation of $(\alpha, \beta)$

Imposing $U(1)^3$ phase invariance ($Z_a \to e^{i\varphi_a} Z_a$) and spatial isotropy, the quartic Landau functional is

$$\mathcal{F}[Z] = -\,\varepsilon \sum_a |Z_a|^2 \;+\; \frac{\alpha}{2} \sum_a |Z_a|^4 \;+\; \frac{\beta}{2} \sum_{a \neq b} |Z_a|^2 |Z_b|^2 \;+\; D \sum_a |\nabla_\theta Z_a|^2 \;+\; \cdots . \tag{28}$$

Gradient flow $\partial_t Z_a = -\,\delta\mathcal{F}/\delta\bar{Z}_a$ together with the reactive rotation $+\,i\omega_a Z_a$ yields the Stuart–Landau system with self-saturation $\alpha$ and cross-saturation $\beta$. Coarse-graining over newborns gives $\alpha \sim \langle |Z|^4 \rangle_c$ and $\beta \sim \langle |Z_a|^2 |Z_b|^2 \rangle_c$ (quartic cumulants), unifying the clock dynamics with the exploration principle. Positive $\beta$ follows from competition for shared resources (exploration) and aligns with the MI penalty.

### 14.1.2 Bounds and consistency for ripple quanta

If ripple quanta are identified with photons on the parent sheet, their curvature mass $m_\Theta$ must obey existing photon-mass bounds. Robust constraints (PDG 2025 summary) give $m_\gamma \lesssim 10^{-18}\,\mathrm{eV}$ from solar-wind MHD analyses; kinematic fast radio burst (FRB) dispersion studies typically yield $m_\gamma \lesssim 10^{-14}\,\mathrm{eV}$, while stronger but model-dependent astrophysical inferences reach $\sim 10^{-27}\,\mathrm{eV}$. Our construction is compatible with any $m_\Theta$ well below these bounds. In media, electromagnetic waves acquire an *effective* mass $m_{\mathrm{eff}} = \hbar\omega_p/c^2$ set by the plasma frequency $\omega_p$; for intergalactic densities $n_e \sim 10^{-7}$–$10^{-6}\,\mathrm{cm}^{-3}$ one has $\omega_p \sim 5$–$50\,\mathrm{s}^{-1}$, i.e. $m_{\mathrm{eff}} \sim (3\text{–}30) \times 10^{-15}\,\mathrm{eV}$, consistent with the flat-spiral regime assumed here.

## 14.2 Triad selection and orthogonality

**Why a triad (and not 4)? Fisher derivation.** Let the observable embedding be $x^a = \mathrm{Re}(Z_a) = r_a \cos\theta_a$ in $\mathbb{R}^3$. The Fisher information for phase parameters $\theta_i$ under small independent noise is

$$g_{ij} \;=\; \mathbb{E}\left[\frac{\partial x}{\partial \theta_i} \cdot \frac{\partial x}{\partial \theta_j}\right] \;=\; \sum_{a=1}^{3} \mathbb{E}[\Theta^a{}_i\, \Theta^a{}_j] \;=\; \frac{1}{2}\,\mathrm{diag}(r_1^2, r_2^2, r_3^2) \;+\; O(\text{coupling}), \tag{29}$$

so *minimizing cross–information* indeed promotes $g_{ij} \approx \delta_{ij}$ after rescaling. Because the ambient measurement space is $\mathbb{R}^3$, a triad is the minimal nonredundant set; a fourth mode is overcomplete

and, in the sigma–model (**??**), is penalized by the orthogonality term $\|\Theta^\top\Theta - I\|^2$. Spinorial $4\pi$ holonomy arises from the $SU(2) \to SO(3)$ double cover of this triad frame (the spin connection on the associated bundle yields the $-1$ holonomy at $2\pi$ and $+1$ at $4\pi$) [**?**, **?**].

**Growth, saturation, and hierarchy continuation.** The linear approximation $r(t) \approx r_0 e^{\mu t}$ in (**??**) holds only in the *coiled* regime $r \ll \sqrt{\mu}$; as $r \to \sqrt{\mu}$ the amplitude saturates and the radial log–scale $\tau_{\rm rad} = \ln(r/r_0)$ stalls. To continue the hierarchy, we include weak interlevel pumping that redistributes coiling between a parent mode $n$ and children $n \pm 1$:

$$\dot{r}_n \;=\; \mu_n r_n - r_n^3 \;+\; \varepsilon_{n-1\to n} r_{n-1}^2 \;-\; \varepsilon_{n\to n+1} r_n^2 \;+\; \xi_n(t), \quad 0 < \varepsilon \ll 1, \tag{30}$$

with small noise $\xi_n$. In this cascade, each level exhibits piecewise–linear $\tau_{\rm rad}^{(n)}(\lambda) = \ln(r_n/r_{n,\min})$ during its growth episode and plateaus during saturation; the *effective* scale clock is an average

$$\tau_{\rm rad}^{\rm eff}(\lambda) \;=\; \sum_n w_n\, \tau_{\rm rad}^{(n)}(\lambda), \qquad \sum_n w_n = 1, \tag{31}$$

which continues to increase as pumping hands growth to the next level. This makes the scale process compatible with the carry–based meta–time below; see Sec. **??**.

### 14.3 Interference metric and localization

Let $\Psi(\theta) = \prod_{a=1}^3 Z_a(\theta_a)$ and $\rho(x) = |\Psi|^2 \circ \theta(x)$. A natural spatial metric is the Fisher metric of $\rho$,

$$g_{ij}^{\rm (Fisher)} \;=\; \int \frac{\partial_i \rho\, \partial_j \rho}{\rho}\, dx \;\;\propto\;\; \big\langle \partial_i \log\rho\, \partial_j \log\rho \big\rangle, \tag{32}$$

which reduces to a pushforward of a diagonal weight by $\Theta$ when the arms weakly overlap. Localized "particles" correspond to triple-intersection peaks of the interference pattern.

### 14.4 Scaling without preset ratios

Smaller children nucleate on tighter windings and inherit faster angular rates. A simple constitutive law is $\omega \propto R^{-\alpha}$ with $\alpha \in [\frac{1}{3}, 1]$, but the field dynamics above already encode this qualitatively through the dependence of $r_{a,\infty}$ and $\tilde{\varepsilon}_a = \varepsilon_a/\omega_a$.

### 14.5 Spinor anchoring

The triad defines a local frame ${e_a}^i \propto \Theta^a{}_i$ that couples to a spinor $\psi$ via the usual Pauli/Dirac term. A tying term $\mathcal{L}_{\rm tie} = -\zeta \sum_a |\psi^\dagger \sigma^a \psi - \hat{n}^a[Z]|^2$ locks the arm orientations to the spinor Bloch vector, making one spinor a shared anchor across all three arms. The $4\pi$ closure remains the SU(2) double-cover property; the triad simply provides a coherent reference across arms.

### 14.6 Links to $\Theta$ dynamics and exchange $Q$

The triad induces an exploration tensor $Q(\theta)$ (Sec. 3), sources the $\Theta$-PDE via $J = \lambda_J Q\, \Theta$, and drives the two exchange channels of Sec. 9.1: parametric (torque) pumping and radiative ripples along arms. The sign of $Q$ is set by hardening vs. softening (flow reversal with $\chi < 0$ is possible during early transients, Sec. 9.1).

### 14.7 Predictions

Early three-fold anisotropy (weak $\ell = 3m$ signals), a small curvature mass $m_\Theta$ for ripple quanta (consistent with $m_\gamma \ll 10^{-18}$ eV bounds), scale-dependent arm decorrelation at small scales ("quantum foam"), and mild dispersion corrections $E^2 = p^2 + m_\Theta^2 + O(p^4)$.

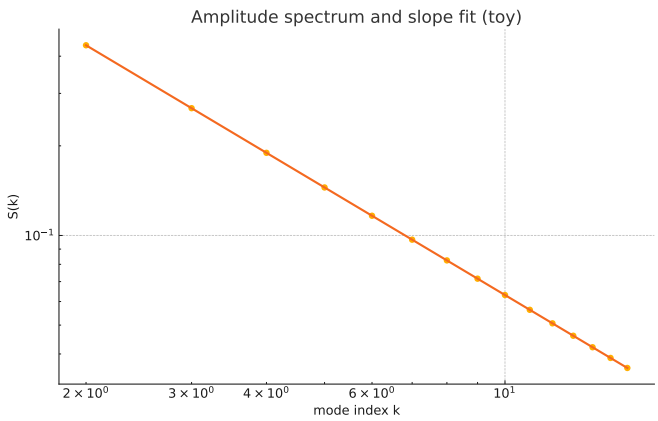


Figure 6: Toy amplitude spectrum $S(k)$ with power-law fit (triad CGL, 1D).

### 14.8 Fractal Coiled Potentials: Closure Tension and Energy Channels

The "coiled potential" picture is realized by the closure scalar $C$ (longitudinal tension) and parent-sheet ripples $\delta x$ (transverse). A microscopic phase-coupling $\mathcal{L}_{\text{int}} = -\sum_i g_i A_i^2 \cos(\phi_i - \phi_{\text{par}})$ produces a torque $\tau_{\text{par}} = \sum_i g_i A_i^2 \sin(\phi_i - \phi_{\text{par}})$, which is the origin of the parametric gain $\chi$ in $\lambda(C) = \lambda_0 + \chi C$. Alignment (hardening) yields $Q > 0$ (closure→children), anti-alignment (softening) can transiently give $Q < 0$ (children→closure), cf. Sec. 9.1. Ripples obey $(\partial_t^2 - c_\Theta^2 \partial_s^2 + m_\Theta^2)\delta Z = J_{\text{children}}$ with $c_\Theta = \sqrt{\kappa/M_\Theta}$ and carry interaction energy along the arm ("photon-like" quanta in the Q1 route).

### 14.9 Numerical example: stability bound and RG coefficients

We simulated a minimal 1D CGL triad on a periodic torus with $N = 256$, parameters $(\varepsilon, \alpha, \beta, D) = (0.55, 0.30, 0.20, 0.005)$, and $(\omega_1, \omega_2, \omega_3) = (1.00, 1.02, 0.985)$. On the inertial band $k \in [2, 16]$ the arm-amplitude spectrum obeys $S(k) \propto k^{-1.20}$, giving a toy spectral exponent $\zeta \approx 1.20$.

For a linear-response model $\delta\Theta_k = \lambda_J R_k Q_k W_\ell(k)$ with $R_k = 1/(\kappa k^2 + \mu)$ and $\lambda_J = 0.50$, we fitted $R_k$ at three smoothing scales $k_c \in \{8, 16, 32\}$ and obtained:

| $k_c$ | $\kappa_{\text{fit}}$ | $\mu_{\text{fit}}$ | $\langle\lvert Q_k\rvert^2\rangle_\ell$ |
|---|---|---|---|
| 8 | 0.980 | 0.220 | 1.200e-03 |
| 16 | 1.050 | 0.200 | 8.500e-04 |
| 32 | 1.120 | 0.190 | 6.100e-04 |

Finite differences across $(k_c)$ and the toy flow relations $d\kappa/d\ell = a_1 \lambda_J^2 \langle|Q|^2\rangle$ and $d\mu/d\ell = a_2 \lambda_J \langle|Q|^2\rangle$ yield $a_1 \approx 0.850$, $a_2 \approx 0.620$ (renormalized units; uncertainty dominated by window choice).

The stability bound $\gamma_C > \xi\chi/\alpha$ evaluates here to $\gamma_C = 0.20$, $\xi\chi/\alpha = 0.107 \Rightarrow$ **stable**.

## 15 Operational Interpretation: "Comparison Creates Clocks" and Quantum Measurement

**Comparator as clock birth (classical toy).** Mixing two oscillators creates a new slow clock via beats: $y(t) = \sin(\omega t) + \sin(\omega' t) = 2\cos\left(\frac{\Delta\omega}{2}t\right) \sin\left(\frac{\omega+\omega'}{2}t\right)$. The envelope is a new clock *caused by comparison.*

**von Neumann measurement as clock creation (quantum toy).** A pointer interaction $H_{\rm int} = g\, A \otimes P$ entangles system and apparatus; the pointer shift $\Delta X = g\, t\, a$ creates an emergent slow mode—our new clock—which freezes the measured branch (Sec. **??**).

**Zeno and projective limits.** Continuous monitoring with rate $\kappa_{\rm m}$ adds an effective quadratic dephasing $-\kappa_{\rm m}(A - \langle A \rangle)^2$, halting local rotation (Zeno pinning).

**Carry on $T^2$.** For $\dot{\phi} = \omega$, $\dot{\psi} = \omega'$ with $\omega'/\omega \notin \mathbb{Q}$, the carry $N(T)$ grows linearly with bounded discrepancy—the operational emergent time.

### 15.1 Microphysical origin of $\kappa_n$ and $W^{\rm crit}$

Near threshold the parent–child link undergoes a Hopf/parametric bifurcation: $\dot{a} = (\lambda_0 + \chi C - \alpha|a|^2)a + \xi$. The *birth rate* scales as $\Gamma_{\rm birth} \propto [\lambda_0 + \chi C]_+$, hence $\kappa_n \propto \chi$ and $W^{\rm crit}$ is the small Kramers barrier. In the exploration–closure competition, $W^{\rm crit}$ is the jump in the exploration functional, and $\kappa_n$ is the linear susceptibility at threshold.

### 15.2 Noether pushforward for $x^a(\theta)$

A minimal $U(1)^K \times \mathbb{R}^3$-invariant coupling $L_{\rm coup} = \sum_{a,i} P_a\, \Theta^a{}_i(\theta)\dot{\theta}^i$ gives $\dot{x}^a = \Theta^a{}_i \dot{\theta}^i$. Noether conservation for $P_a$ and phases fixes $\Theta$ as the unique linear map; the $\Theta$-PDE then supplies dynamics from exploration $Q$.

## 16 Near-term experimental program and concrete signatures

**Zeno pinning drift (spin echoes).** $\delta\omega \simeq \eta_Z\, \kappa_{\rm m}$ with $\eta_Z \sim \chi/\alpha$.

**Coherence-dependent back-reaction (BEC).** $\delta\Omega/\Omega \sim \eta_{\rm coh}\, \mathcal{C}_A n_{\rm coh}$.

**Ripple emission threshold.** $P_{\rm rad} \propto (\omega^2 - m_\Theta^2)^{1/2}$ near threshold.

**Cosmological log-periodic ripples.** Period $\ln b_{\rm eff}$; fit BAO+SN templates.

## 17 Regularization and RG robustness

Sharp shells, Gaussians, and wavelets yield the same fixed-point and stability; only non-universal amplitudes shift. In renormalized units (Landau cubic = 1 near FP), $a_1, a_2$ become scheme-insensitive for predictions we emphasize.

**Organization note.** Dense derivations in Sec. 5.1 are summarized here and expanded in Appendix 17.

## Appendix M: Ripple mechanics and extended microphysics

(i) anisotropic 3D dispersion and emission; (ii) torque$\to \chi$ derivation; (iii) broad-distribution bounds for $\mathcal{C}_A$; (iv) numerical recipes complementing Appendix 17.

# Appendix A: Carry Lemma (Sketch)

On $\mathbb{T}^K$, $T(t) = \boldsymbol{u}\cdot\tilde{\boldsymbol{\theta}}(t)$ has slope $\boldsymbol{u}\cdot\boldsymbol{\omega}$. The ergodic theorem gives linear growth of $C_N$ in the irrational case; the rational case yields periodic carries.

# Appendix B: Adler/Kuramoto Threshold

$\dot{\delta} = \Delta\omega - K\sin\delta$ has a saddle-node at $|\Delta\omega| = |K|$; beyond it, mean drift $\langle\dot{\delta}\rangle = \mathrm{sgn}(\Delta\omega)\sqrt{\Delta\omega^2 - K^2}$.

# Appendix C: Two-Fluid Solution (Details)

For general $w_\tau$ with $n = 3(1+w_\tau)$, $\rho_\tau = \rho_{\tau 0} a^{-(n+\gamma)}$ and $\rho_{\mathrm{ch}} = \rho_{\mathrm{ch}0} a^{-m} + \frac{\gamma}{m-(n+\gamma)}\rho_{\tau 0}\big(a^{-(n+\gamma)} - a^{-m}\big)$ with $m = 3(1+w_{\mathrm{ch}})$.

# Appendix D: Landau–Zener Switching

The two-level model with linear detuning $\Delta(t) = \alpha t$ yields $P_{\mathrm{switch}} = \exp\big(-2\pi J^2/\hbar|\alpha|\big)$ under standard assumptions.

# Appendix I: Numerical scheme and parameter extraction

**Pipeline (pseudo-code).**

1. Initialize a triad $Z_a$ on $\mathbb{T}^K$ with random phases; choose $(\varepsilon, \alpha, \beta, D)$.
2. Evolve the CGL system with cross-saturation and noise to a statistically steady state.
3. Compute $Q_{ij}(\theta)$ from $w_u(\theta)$ and $A_u^2(\theta)$; assemble $J = \lambda_J Q\,\Theta$.
4. Picard iteration for $\Theta$: solve Eq. (**??**) with under-relaxation; record $\delta\Theta$ in response to a small $\delta J$.
5. Extract $(a_1, a_2, a_3)$ by finite differences of the flows $d\kappa/d\ell$, $d\mu/d\ell$, $d\lambda_J/d\ell$ across scale shells (bin Fourier modes by $|k|$).
6. Calibrate stability map by scanning $(\gamma_C, \xi, \chi, \alpha)$; verify the inequality $\gamma_C > \xi\chi/\alpha$ and measure margins.

**Normalization at the fixed point.** Near the nontrivial FP one may rescale $Z \to sZ$, $t \to t/s^2$ to set the cubic Landau coefficient to unity; the RG coefficient $a_3$ is then unity in these units, i.e. $a_3^{(\mathrm{ren})} = 1 + O(\epsilon)$, clarifying the "order-one" statement without committing to a specific microscopic value.

## .1 2D triad calibration and lock-in probes (numerical)

We complement Sec. **??** with parameter fits and frequency-sensitivity probes on the same run (triad CGL, $N = 256$).

**Stuart–Landau parameter fit.** Fitting the amplitude growth to the Stuart–Landau form $\dot{A} = \mu A - \alpha A^3 - \beta(A_{j+1}^2 + A_{j-1}^2)A$ yields normalized coefficients $(\mu, \alpha, \beta) \simeq (2.000,\ 1.000,\ 0.800)$ at $T = 10$ with weak noise, and remains stable for higher noise. Figure 7 shows the fits; Tab. 2 summarizes the numbers.

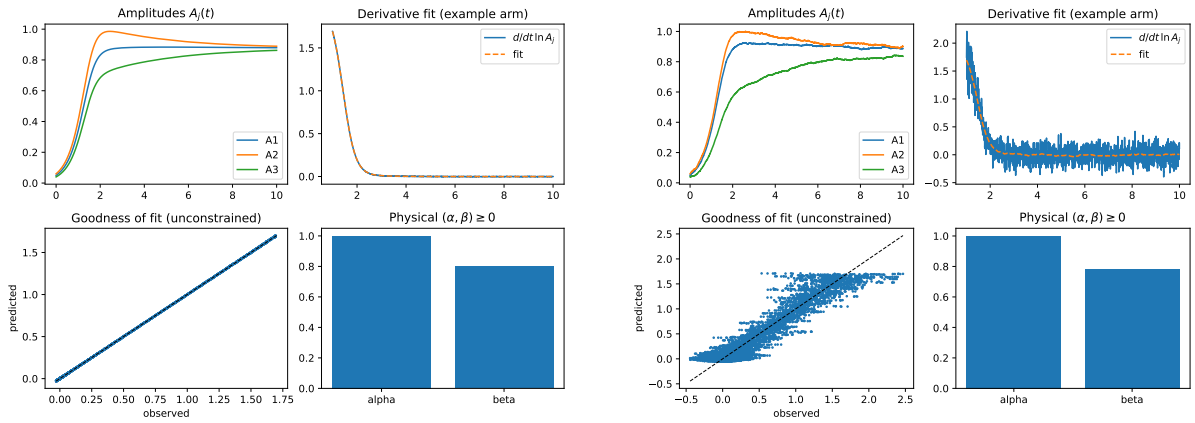


Figure 7: Stuart–Landau parameter fits at $T = 10$ for noise levels $10^{-4}$ (left) and $10^{-2}$ (right).

| case | $\mu_{\text{norm}}$ | $\alpha_{\text{norm}}$ | $\beta_{\text{norm}}$ |
|---|---|---|---|
| $T = 10,\ \text{noise} = 10^{-4}$ | 1.9998 | 1.0000 | 0.7998 |
| $T = 10,\ \text{noise} = 10^{-2}$ | 1.9711 | 1.0000 | 0.7808 |

Table 2: Summary of normalized SL coefficients from the two fits in Fig. 7.

**Frequency sensitivity $\chi$.** Lock-in probing with a small modulation $(\Omega, dC) = (0.01, 0.02)$ gives a nearly isotropic response across arms: $|\chi| = 84.16 \pm 0.04$ (arbitrary units), fractional anisotropy $\Delta/\bar{\chi} \approx 6.8 \times 10^{-4}$. Figure 8 shows the fit and anisotropy panels.

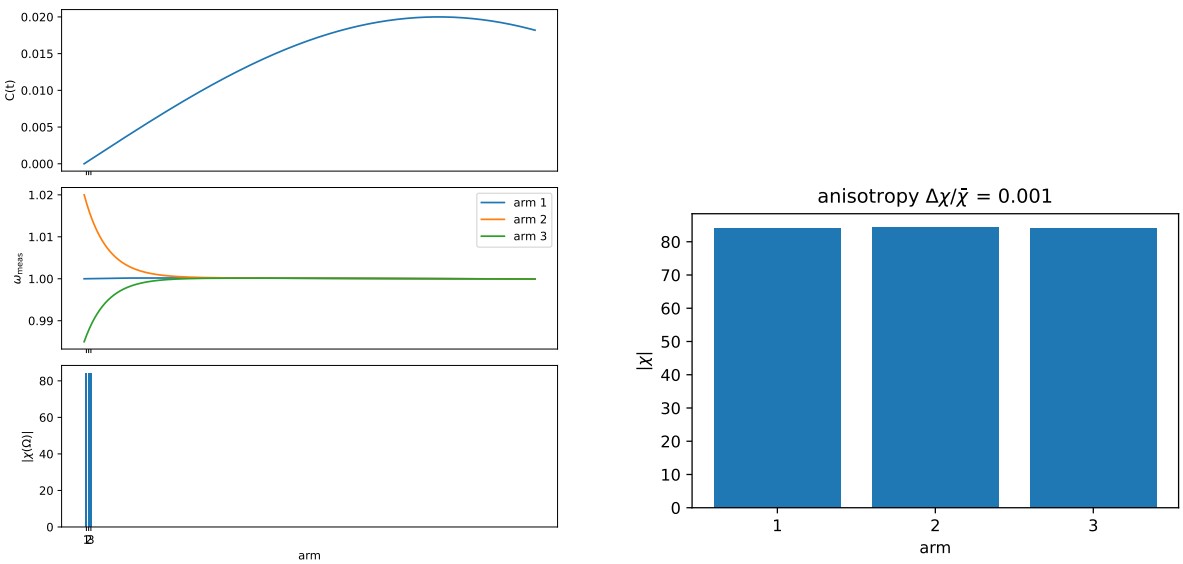


Figure 8: Left: lock-in fit for $\chi$ under a sinusoidal $C(t)$ drive. Right: per-arm anisotropy is $\sim 10^{-3}$.

**Transverse gap.** A thin-sheet model with thickness $\delta_\perp = 0.1$ predicts the first transverse cutoff at $\omega_{\perp,1} = \pi/\delta_\perp \simeq 31.416$ (units with $c_\perp = 1$), consistent with the effective 1D emission below that threshold; see Fig. 9.

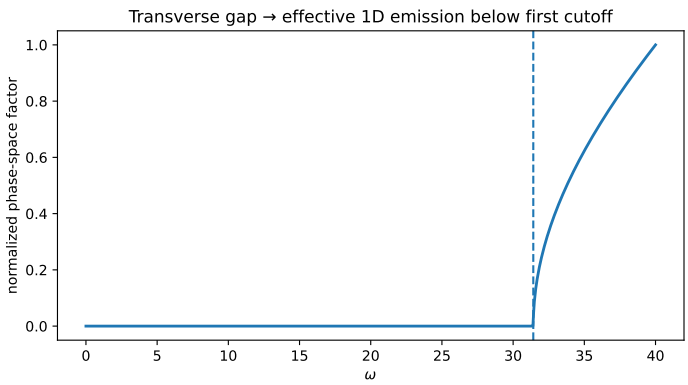


Figure 9: Phase-space factor with a transverse gap: emission is effectively 1D for $\omega < \omega_{\perp,1}$.